Isaac Watts

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Professional Summary

Graduating summa cum laude with a Bachelor of Science in Computer Science and Mathematics from Virginia State University, I bring a strong foundation in software engineering principles, algorithms, and systems design. My academic and internship experiences have equipped me with hands-on skills in developing, testing, and deploying software applications, as well as collaborating in agile team-based engineering environments. I have also contributed to research in machine learning and robotics, further showcasing my ability to write clean, efficient code. Outside of academics, I pursue logic-based activities such as chess, puzzles, and strategy games which reinforce my ability to think critically, solve problems efficiently, and continuously improve my technical acumen.

Technical Skills

Programming Languages: Python, C, C++, Java, C#, Kotlin, MATLAB, LaTeX

Databases: Relational databases (MySQL, PostgreSQL, and SQLite) and Nonrelational databases (MongoDB)

Data Processing & BI Tools: Tableau, Power BI, Excel, Pandas, NumPy, Matplotlib, PyTorch

Version Control & Containerization: Git, GitLab, Docker

Operating Systems: Windows 11 and Ubuntu 22.04

Experience

Robotics Student Researcher — *Adelphi, MD*
Army Research Lab — *May 2024 – August 2024*

- Trained an AI agent using Unity observations to re-establish radio communications through reinforcement learning by simulating radio waves with C# via Unity Machine Learning Agents Toolkit (ML-agents) and the library PyTorch.

Tutor — *Petersburg, VA*
Virginia State University — *September 2023 – May 2024*

- Tutored peers in SQL, R, Python, and data-focused subjects such as discrete math, calculus, and statistics.

Robotics Student Researcher — *Adelphi, MD*
Army Research Lab — *May 2023 – August 2023*

- Developed an image processing tool to convert an 8-bit image to a 16-bit format. Compressed that 16-bit image using an image transport layer that allowed the data to be passed through the network using C++ (ROSCPP).
- Automated writing Robot Visualization files for multi-robot experiments by using a Python (ROSPY) automation script, resulting in no longer needing to manually do everything in an RVIZ GUI or file, boosting productivity.

Education

Virginia State University — *GPA: 3.9*
Bachelor's Degree in Computer Science

- **Coursework:** Intro to Data Science, Probability & Statistics for CS, and Digital Image Processing

Virginia State University — *GPA: 3.9*
Bachelor's Degree in Mathematics

- **Coursework:** Linear Algebra and Discrete Math

John Tyler Community College — *GPA: 4.0*
Associates in General Studies

Projects

Inverse Problem Modeling and Machine Learning *github.com/IwattsX/med_bodies*

- Simulated voltage readings from a circular mesh with an embedded inclusion using MATLAB.
- Trained a neural network to localize the inclusion based on voltage patterns using Python (ELU + MSE loss + Adam optimizer) using PyTorch.
- Reconstructed 2D images of the mesh to visualize inclusion location from the neural net's predictions.
- Combined physical modeling and machine learning to solve an inverse imaging problem.

Steam Scouts *github.com/IwattsX/Digi-Dynamics*

- Created a MySQL-backed Django web app to collect and explore Steam game metadata from Steam's web API.
- Parsed and stored structured data (e.g., game type, price, DLC) using Python and SQL connector.
- Enabled filtering and querying by game type, release date, and content type.

Robotic Arm Simulation using Isaac Sim *github.com/stars/IwattsX/lists/robotsim*

- Created a ROS2 node using rclpy (Python) that publishes Joint State data using Omniverse's Isaac Sim and Isaac Lab, resulting in a ROS2 node that consistently publishes the correct Joint State data of the robot in the simulation.
- Graphed hand data in Matplotlib that was generated from an Oculus Quest alongside Unity's game engine.
- Created and modified URDF files of a robot hand to visualize and control the joints.

Data Visualizations from CDC dataset *github.com/IwattsX/IntroToDS*

- Cleaned data and filtered out null values directly within Tableau to ensure accurate visualization.
- Built a multi-page dashboard exploring trends in chronic diseases by state (mainly Virginia), gender, and demographics.
- Leveraged Tableau's storytelling tools to highlight key insights on health disparities and risk factors.
- Designed interactive elements (filters, highlighters, parameter controls) for user-driven exploration.
- https://public.tableau.com/app/profile/isaac.watts/viz/Disease_indicators_and_insights_IW/Story1

Certificates

- **Getting Started with Data by IBM Skillsbuild** July 2025
- **ETL in Python and SQL** June 2025
- **Anaconda Python for Data Science Professional Certificate** May 2025
- **Learning Python (2021)** May 2025
- **Statistics Foundations 1: The Basics** May 2025
- **Statistics Foundations 3: Using Data Sets** May 2025
- **Introduction to Data Science** May 2025
- **Learning Excel: Data Analysis** May 2025
- **MATLAB Programming Techniques** March 2025
- **MATLAB Fundamentals** February 2025
- **Intermediate Android Development Course for Virginia State University** December 2024

Publications

A Transformer Approach for Camera-to-LIDAR Data Registration October 2024

Ju Wang, Yong Tang, Venkat R. Dasari, Billy Geerhart, Brian Rapp, Peng Wang, Wei-Bang Chen, ***Isaac Watts***,

10.1109/IRI62200.2024.00072